

AI Security

Trustworthy ML

Sangdon Park

POSTECH

May 26, 2026

Outline

1 Introduction

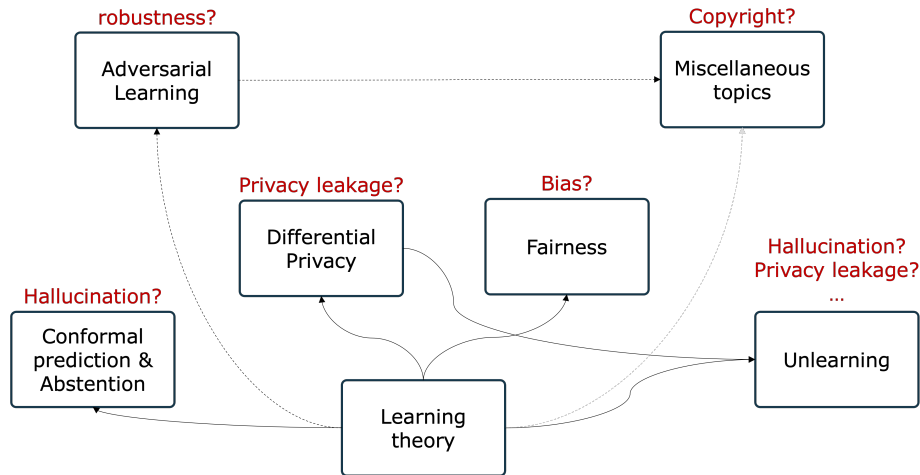
2 Conformal Prediction

- Offline Conformal Prediction
- Online Conformal Prediction with Full Feedback
- Online Conformal Prediction with Partial Feedback

3 Conformal Abstention

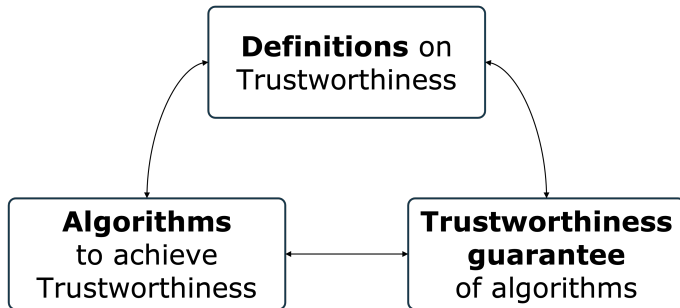
- Offline Conformal Abstention
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Trustworthy Machine Learning: Overview



- Check out our Trustworthy ML class (AIGS703L / CSED703L)

Trustworthy Machine Learning: Key Components



Hallucination in AI Security

- Goal: Learning to minimize hallucination under adversarial attacks

$$\min_f \max_{\delta} \mathbb{E}_x \text{hallucination}(f(x + \delta))$$

- ▶ Learner's Goal: minimize hallucination
- ▶ Adversary's Goal: maximize hallucination

Hallucination in AI Security

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 - ▶ Adversary's Goal: maximize hallucination
- Example in Agentic AI Security:
 - ▶ Goal: Learning to minimize a task failure rate and harmfulness under adversarial attacks

$$\min_{\pi} \mathbb{E}_x [\text{failure}(\pi(x + \delta))] \quad \text{subj. to} \quad \mathbb{E}_x [\text{harmfulness}(\pi(x + \delta))] \leq \alpha \quad \forall \delta$$

Hallucination in AI Security

- Goal: Learning to minimize hallucination under adversarial attacks

$$\min_f \max_{\delta} \mathbb{E}_x \text{ hallucination}(f(x + \delta))$$

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- Research questions:
 - ① How to control a metric (e.g., hallucination) without adversaries?
 - ② How to control a metric (e.g., hallucination) with adversaries? – open question

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Conformal Prediction



Vladimir Vovk

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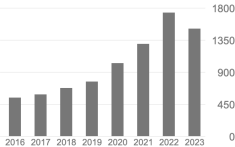
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TITLE	CITED BY	YEAR
Algorithmic learning in a random world V Vovk, A Gammernan, G Shafer Springer	1435	2005
Ridge regression learning algorithm in dual variables C Saunders, A Gammernan, V Vovk	1044	1998
Aggregating strategies V Vovk Proceedings of 3rd Annu. Workshop on Comput. Learning Theory, 371-383	956	1990
A Tutorial on Conformal Prediction. G Shafer, V Vovk Journal of Machine Learning Research 9 (3)	853	2008

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Conformal Prediction



Vladimir Vovk

Algorithmic learning in a random world

Authors [Vladimir Vovk, Alexander Gammerman, Glenn Shafer](#)

Publication date [2005/3/1](#)

Volume [29](#)

Publisher [Springer](#)

Description This book is about conformal prediction, an approach to prediction that originated in machine learning in the late 1990s. The main feature of conformal prediction is the principled treatment of the reliability of predictions. The prediction algorithms described—conformal predictors—are provably valid in the sense that they evaluate the reliability of their own predictions in a way that is neither over-pessimistic nor over-optimistic (the latter being especially dangerous). The approach is still flexible enough to incorporate most of the existing powerful methods of machine learning. The book covers both key conformal predictors and the mathematical analysis of their properties.

Algorithmic Learning in a Random World contains, in addition to proofs of validity, results about the efficiency of conformal predictors. The only assumption required for validity is that of "randomness"(the prediction algorithm is presented with ...

Total citations [Cited by 1435](#)



Scholar articles [Algorithmic learning in a random world](#)
[V Vovk, A Gammerman, G Shafer - 2005](#)
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...we are *hedging* the prediction — we are adding to it a statement about how strongly we believe it.
— Vovk et al., 2005

Motivation

Conventional prediction:

$$\hat{y} \left(\text{Toy Terrier} \right) = \widehat{\text{Bulldog}}$$

Conformal prediction:

$$\hat{c} \left(\text{Toy Terrier} \right) = \left\{ \begin{array}{l} \text{Toy terrier} \\ \text{Bulldog} \\ \text{poodle} \end{array} \right\}$$

Motivation

Conventional prediction:

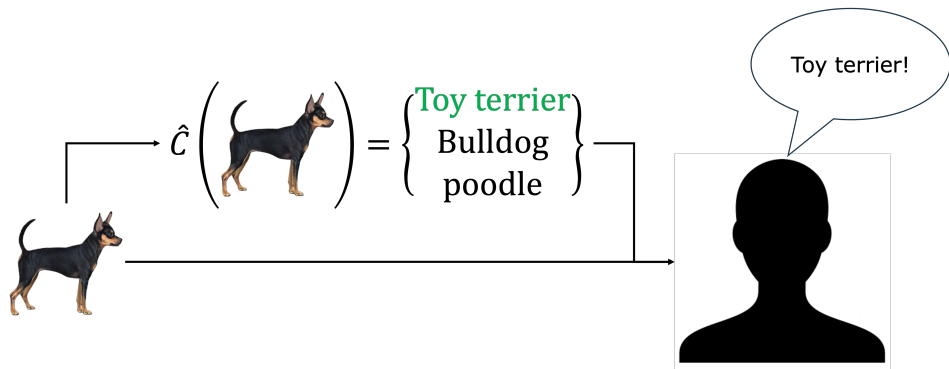
$$f : \mathcal{X} \mapsto \mathcal{Y}$$

Conformal prediction:

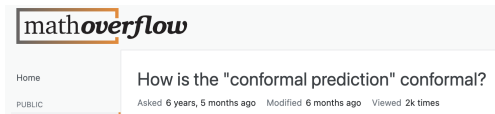
$$C : \mathcal{X} \mapsto 2^{\mathcal{Y}}$$

- Conventional prediction is a “point” prediction.
- Conformal prediction is a set-valued prediction.
- The set contains “likely-correct” alternative options.
 - ▶ The set size measures “uncertainty”!
- Why not confidence prediction? User-friendly?

Motivation: Decision Making



Why “Conformal”?



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+100



Thanks for your interest. The term “conformal prediction” was suggested by Glenn Shafer, and at first I did not like it exactly for the reason that you mention: it has nothing (or very little) to do with conformal mappings in complex analysis. But then I discovered other meanings, even in maths; e.g., Wikipedia has five on its disambiguation page for “conformal”:

- Conformal film on a surface (same thickness)
- Conformal fuel tanks on military aircraft
- Conformal coating in electronics
- Conformal hypergraph, in mathematics
- Conformal software, in ASIC Software

So the word did not look taken to me anymore. The expression that we had used before Glenn proposed “conformal prediction” was even worse (“transductive confidence machine”).

Thanks to Hengrui Luo for drawing my attention to this question.

As for question (2), the answer depends on which robust predictors you have in mind. The predictors with most similar properties are the ones in classical statistics (such as the standard prediction intervals in linear regression based on Student's t distribution); the main difference is that they are parametric. There is a predictive version of tolerance intervals in nonparametric statistics, but their treatment of objects (x parts of observations (x,y) , where y are labels) is limited. Upper bounds on the probability of error given by standard PAC predictors are often too high to be useful.

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answered Apr 13, 2017 at 7:35



Vladimir Vovk

Conformal (Prediction) Sets

Definition (conformal set)

$$C(x) := \{y \in \mathcal{Y} \mid f(x, y) \geq q\}$$

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- We are using more recent notations based on inductive conformal prediction.
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 - ▶ Note that inductive conformal prediction [Papadopoulos et al., 2002] is an efficient variation of full conformal prediction [Vovk et al., 2005].

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- $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$: a conformity scoring function
 - ▶ Measures how well (x, y) *conforms* to a trained model f (via a proper training set)
 - ▶ $f(x, y)$ is a likelihood of x for being y

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 - ▶ $f(x, y)$ is a likelihood of x for being y
- q : A parameter to be chosen by an algorithm.

Conformity Scoring Functions I

Conformity scoring functions can be (almost) any model!

Example (classification)

$$f(x, y) := f_{\text{cls}}(x, y)$$

- f_{cls} : a classification model, e.g., resnet

Conformity Scoring Functions II

Conformity scoring functions can be (almost) any model!

Example (standard regression in 1-dimension)

$$f(x, y) := -|\mu(x) - y|$$

- μ : a regressor

Back to Conformal Sets

Definition (conformal sets)

$$C(x) := \{y \in \mathcal{Y} \mid f(x, y) \geq q\}$$

- A conformity scoring function f is given.
- f is a target to measure uncertainty.
- How to choose q ?

Assumption: Exchangeability

Assumption

A sequence of random variables $X_1, X_2, \dots$ is exchangeable if for any permutation σ , the following holds:

$$\mathbb{P} \{X_1 = x_1, X_2 = x_2, \dots\} = \mathbb{P} \{X_{\sigma(1)} = x_1, X_{\sigma(2)} = x_2, \dots\}.$$

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- The i.i.d. assumption implies the exchangeability assumption (why?).

A Goodness Metric: Coverage Guarantee

Definition (coverage guarantee)

$$\mathbb{P}\left\{Y_{n+1} \in \hat{C}(X_{n+1})\right\} \geq 1 - \alpha$$

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- (X_i, Y_i) for $i = 1, \dots, n + 1$: the exchangeable samples (thus the i.i.d. samples)

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- $\hat{C}$: A conformal set constructed by the training set

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- (X_i, Y_i) for $i = 1, \dots, n + 1$: the exchangeable samples (thus the i.i.d. samples)
- $\hat{C}$: A conformal set constructed by the training set
- $1 - \alpha \in (0, 1)$: A desired coverage rate

Quantile

Quantile of a Distribution

The level β quantile of a distribution F :

Definition (quantile)

$$\text{Quantile}(\beta; F) := \inf\{z \mid \mathbb{P}\{Z \leq z\} \geq \beta\}$$

- F : a distribution over the augmented real line, $\mathbb{R} \cup \{\infty\}$
- $Z \sim F$
 - ▶ allows multiple instances of the same element

Quantile

Quantile of an Empirical Distribution

The level β quantile of an empirical distribution of the values $v_{1:n}$:

Definition (quantile)

$$\text{Quantile}(\beta; v_{1:n}) := \text{Quantile} \left(\beta; \frac{1}{n} \sum_{i=1}^n \delta_{v_i} \right)$$

- $v_{1:n} := \{v_1, \dots, v_n\}$: an unordered multiset
- δ_a : a δ -distribution (*i.e.*, a point mass at a)

Quantile Algorithm

Definition (quantile algorithm)

Given $(X_1, Y_1), \dots, (X_n, Y_n)$,

$$\hat{q}_{1-\alpha} := \text{Quantile}(1 - \alpha, V_{1:n} \cup \{\infty\}),$$

where $V_i := -f(X_i, Y_i)$.

- The implementation is as simple as finding the k -th smallest value.

Coverage Guarantee of the Quantile Algorithm

Theorem ([Vovk et al., 2005, Lei et al., 2018])

Assume that (X_i, Y_i) for $i \in \{1, \dots, n+1\}$ are exchangeable. For any scoring function f and any $\alpha \in (0, 1)$, denote the conformal set by

$$\hat{C}(x) := \left\{ y \in \mathcal{Y} \mid -f(x, y) \leq \hat{q}_{1-\alpha} \right\}.$$

Then, we have

$$\mathbb{P} \left\{ Y_{n+1} \in \hat{C}(X_{n+1}) \right\} \geq 1 - \alpha,$$

where the probability is taken over (X_i, Y_i) .

- This is a marginal coverage guarantee.

Power of Conformal Prediction

The coverage guarantee is drawn with minimal assumptions.

- It does not make assumptions on a distribution except for the exchangeability.
- The guarantee holds for any conformity scoring function.

Size of Conformal Sets

- Application-dependent issues
 - ▶ classification: set size
 - ▶ 1-D regression: interval length
 - ▶ multi-dimensional regression: *e.g.*, volume
- Larger set: uncertain (*e.g.*, the entire set)
- Smaller set: more certain (*e.g.*, a singleton)

Conclusion

- Conformal prediction is a powerful tool to construct a prediction set (for measuring uncertainty) with correctness guarantees.
- Conformal prediction has many applications due to its “distribution-free” and “scoring-function-free” nature.

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Motivation: Distribution Shift

- The main assumption of conformal prediction: exchangeability or i.i.d.
- In practice, this is fragile due to distribution shifts.
- Type of distribution shifts
 - ▶ Covariate shift
 - ▶ Label shift
 - ▶ ...
 - ▶ Adversarial shift

Adversarial Shift

- Learning setup: follows an online learning setup, *i.e.*,
 - ▶ there are multiple shifts over time
 - ▶ $p_t(x, y)$: a distribution at time t
 - ▶ $(x_t, y_t) \sim p_t(x, y)$: a labeled example sampled at time t

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 - ▶ $(x_t, y_t) \sim p_t(x, y)$: a labeled example sampled at time t
- Assumption: no restriction on shifts
- Conformal prediction under distribution shift
 - ▶ [Gibbs and Candès, 2021]: provides the coverage guarantee
 - ▶ [Bastani et al., 2022]: provides the coverage guarantee for fairness

Adaptive Conformal Prediction

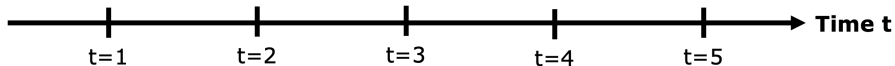
Can we learn conformal sets under distribution shift?

Setup:

- $\mathcal{X}$: example space
- $\mathcal{Y}$: label space
- $C_t : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$: a conformal set
- A learning game between a learner and nature
 - for** $t = 1, \dots, T$ **do**
 - Learner receives an example $x_t \in \mathcal{X}$
 - Learner outputs a *conformal set* $C_t(x_t) \in 2^{\mathcal{Y}}$
 - Learner receives a true label $y_t \in \mathcal{Y}$
 - Learner suffers loss $\mathbb{1}(y_t \notin C_t(x_t))$
 - Learner update a *parameter of a conformal set*
 - end for**

Adaptive Conformal Prediction

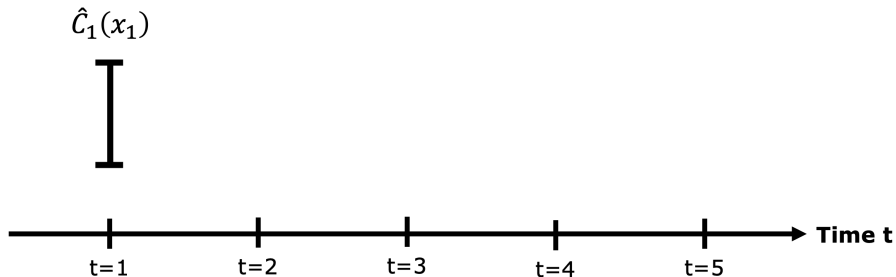
Intuition



- Adaptive conformal prediction progressively adjust a prediction set such that it covers a desired number of samples.

Adaptive Conformal Prediction

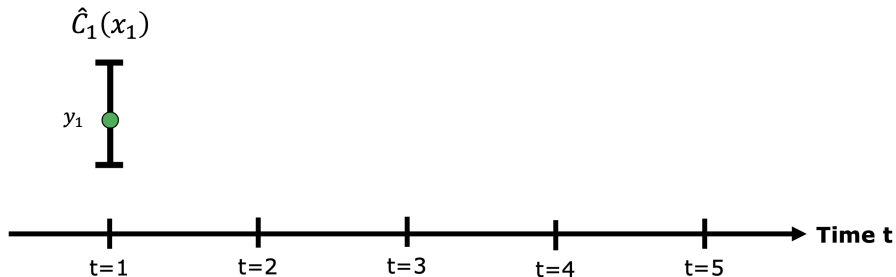
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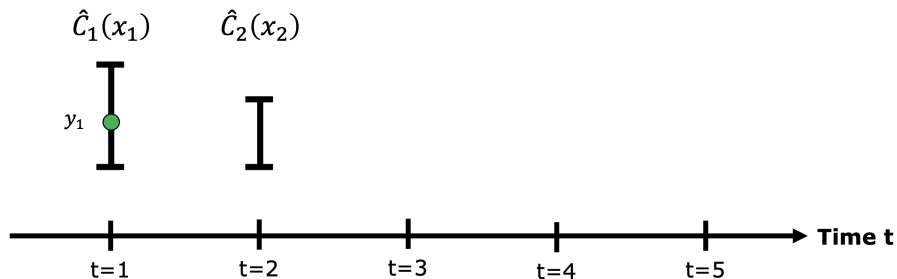
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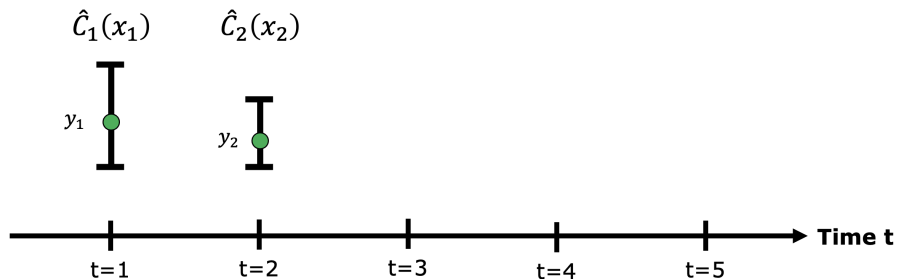
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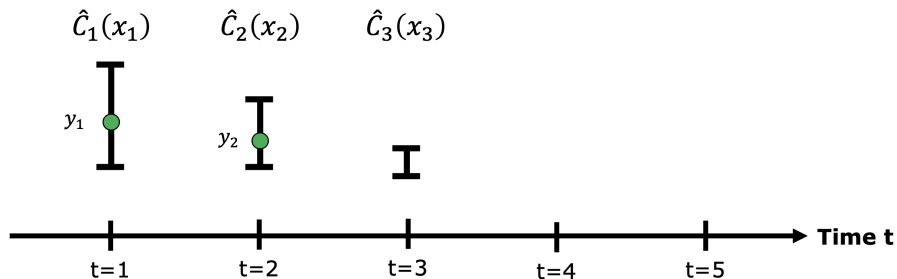
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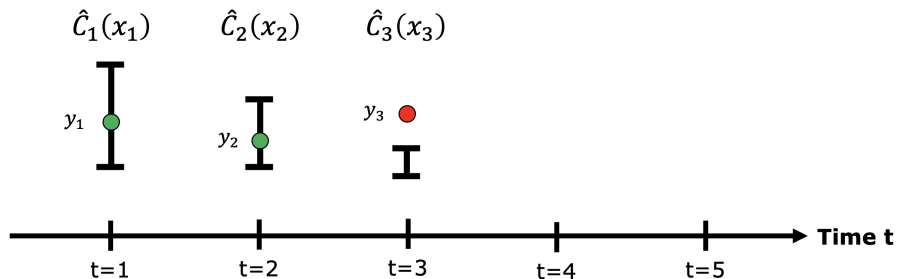
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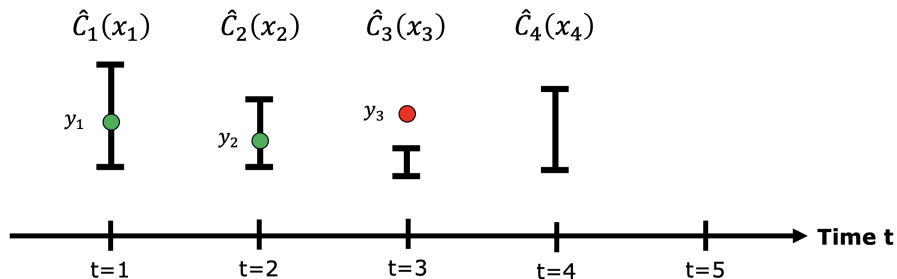
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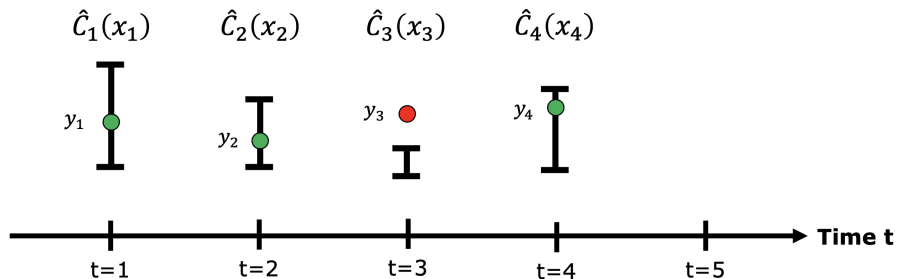
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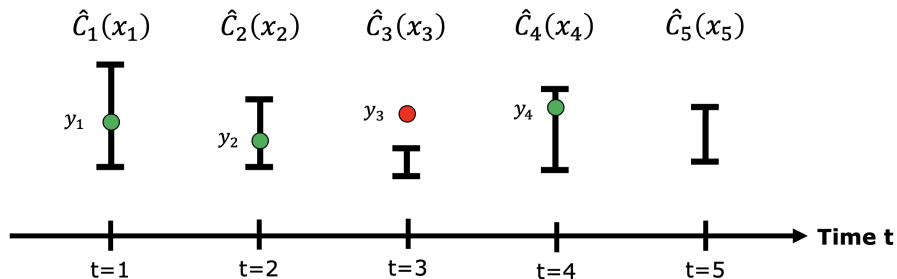
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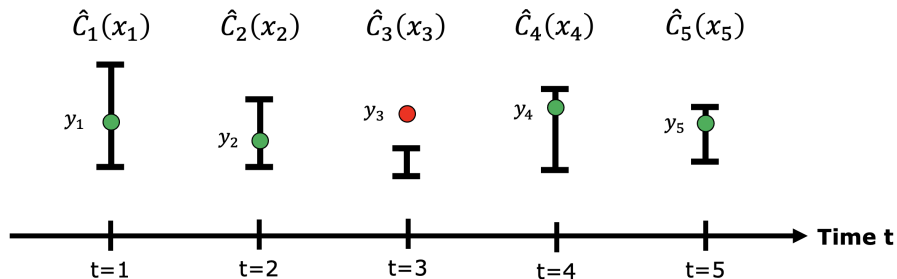
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A Goodness Metric: “Empirical” Coverage Guarantee

Definition (empirical coverage guarantee)

$$\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1} \left(y_t \notin \hat{C}_t(x_t) \right) - \alpha \right|$$

- $1 - \alpha$: a desired coverage rate
- T : a time horizon
- $\hat{C}_t$: a conformal set at time t constructed by an algorithm
- It is similar to the regret definition (but not exactly the same).
- We wish to bound this quantity.

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- We wish to bound this quantity.
- Why not use the PAC guarantee?
 - ▶ the PAC guarantee is for the batch learning.

Algorithm

Main Ideas

- Run the batch conformal prediction (CP) for each time
- But adjust the coverage α for the batch CP to satisfy the empirical coverage guarantee.

Algorithm

Algorithm 1 A standard version of Adaptive Conformal Inference [Gibbs and Candès, 2021]

```
1:  $t_1 \in \{1, \dots, T\}$ 
2:  $\alpha_{t_1} \in [0, 1]$ 
3: for  $t = t_1, \dots, T$  do
4:    $(\mathcal{D}_{\text{train}}^{(t)}, \mathcal{D}_{\text{cal}}^{(t)}) \leftarrow$  Randomly split the data  $\{(x_i, y_i)\}_{i=1}^{t-1}$  and obtain non-conformity scores
5:    $S_t \leftarrow$  Update a scoring function using  $\mathcal{D}_{\text{train}}^{(t)}$ 
6:    $q_t \leftarrow \text{Quantile}(1 - \alpha_t, \mathcal{D}_{\text{cal}}^{(t)} \cup \{\infty\})$ 
7:   Observe  $x_t$ 
8:   Predict  $\hat{C}_t(x_t)$ 
9:   Observe  $y_t$ 
10:  Update  $\alpha_{t+1} \leftarrow \alpha_t + \gamma \left( \alpha - \mathbb{1} \left( y_t \notin \hat{C}_t(x_t) \right) \right)$ 
11: end for
```

- A conformal set: $\hat{C}_t(x_t) := \{y \in \mathcal{Y} \mid S_t(x_t, y) \leq q_t\}$
- Until t_1 , the algorithm simply collects data.
- The algorithm is not randomized.

Coverage Bound

Theorem

For all $T \in \mathbb{N}$, $\alpha \in (0, 1)$, and $\gamma > 0$,

$$\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1} \left(y_t \notin \hat{C}_t(x_t) \right) - \alpha \right| \leq \frac{\max\{\alpha_1, 1 - \alpha_1\} + \gamma}{T\gamma}$$

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$$\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1} \left(y_t \notin \hat{C}_t(x_t) \right) - \alpha \right| \leq \frac{\max\{\alpha, 1 - \alpha\} + \gamma}{T\gamma}$$

- The coverage decreases by $\mathcal{O}\left(\frac{1}{T}\right)$

Coverage Bound

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- This holds for any sequence $((x_1, y_1), \dots, (x_T, y_T))$!
 - ▶ If $\hat{C}_t(x_t) = \mathcal{Y}$, the adversary will never win without randomization.
- Suppose $\alpha_1 = 0$, $\gamma = 0.01$, and $\varepsilon = 0.01$ (which denotes the upper bound). Then, we $T = 10, 100$ observations to make the empirical coverage close to a desired coverage.

Conclusion

- Adaptive Conformal Inference [Gibbs and Candès, 2021] is the first approach to learn a conformal set under distribution shifts.
- This is an example of running a batch algorithm within an online algorithm.
 - ▶ The time and memory complexity is linear in T .
 - ▶ See a more efficient (and general) approach [Bastani et al., 2022]

Outline

1 Introduction

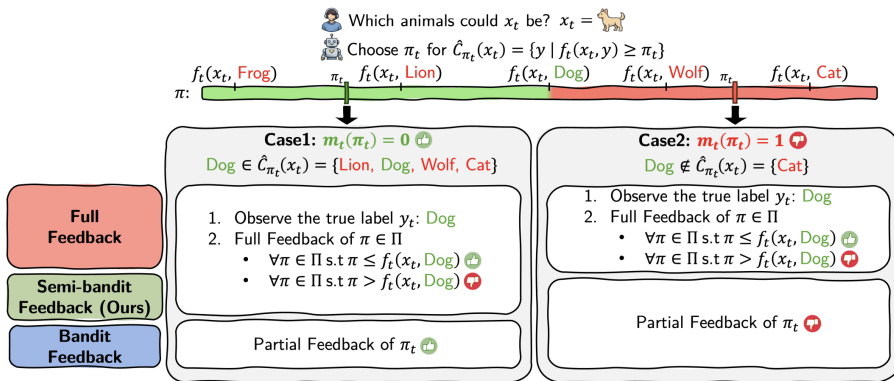
2 Conformal Prediction

- Offline Conformal Prediction
- Online Conformal Prediction with Full Feedback
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OCP with Adversarial Semi-Bandit Feedback [Yang et al., 2026]



- A novel online conformal prediction method, updated by only using partial feedback

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